



# OpenMVC: An Open-Source Library for Learning-based Multi-view Compression

Huiming Zheng

hmzheng@stu.pku.edu.cn

Guangdong Provincial Key Laboratory of Ultra High Definition Immersive Media Technology, Shenzhen Graduate School, Peking University  
Shenzhen, China

Wei Gao\*

gaowei262@pku.edu.cn

Guangdong Provincial Key Laboratory of Ultra High Definition Immersive Media Technology, Shenzhen Graduate School, Peking University  
Peng Cheng Laboratory  
Shenzhen, China

## Abstract

Amidst the swift advancement of 3D vision technology, Multi-view Compression (MVC) has become a crucial technique, widely applied in fields such as virtual reality, augmented reality, autonomous driving, telemedicine, and security surveillance. The technology effectively handles views from multiple cameras, utilizing the inter-view correlations to compress data efficiently. It substantially decreases the data transmission and storage requirements, enabling a richer and more realistic visual experience within the same bandwidth constraints. To further enhance compression performance, new methods continue to emerge. However, the absence of a unified benchmark testing library capable of effectively evaluating existing algorithms poses significant challenges to the further development of the field and the practical deployment of algorithms. To address this issue, we introduce OpenMVC, an Open-Source Library for Learning-based Multi-view Compression. We provide a comprehensive description and analysis of the performance advantages of existing algorithms. Furthermore, we conduct extensive and comprehensive benchmark testing of nine typical algorithms in the last five years, evaluating them in a consistent environment across various metrics. The open-source library for OpenMVC is available at <https://openi.pcl.ac.cn/OpenAICoding/OpenMVC>.

## CCS Concepts

• Theory of computation → Data compression.

## Keywords

Multi-view Compression, Open-source Software, Deep Learning

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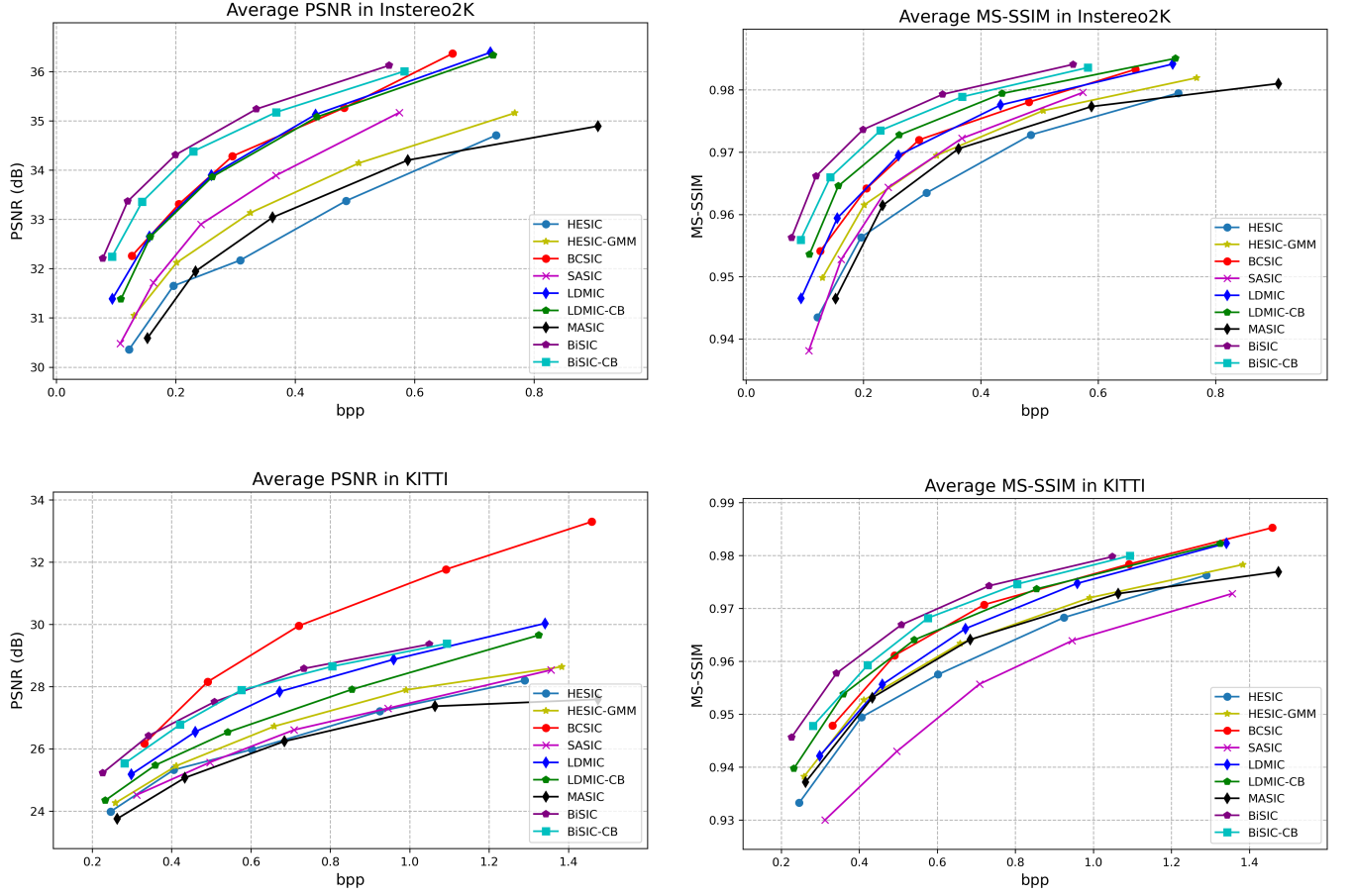
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## 1 Introduction

In order to describe objects and scenes in the three-dimensional world more realistically and vividly, it is typically necessary for cameras to capture from different angles and perspectives. Multi-view images are crucial for domains such as 3D reconstruction [10, 26], virtual reality [6, 16], point cloud processing [27–29], and scene monitoring [1, 15]. However, multi-view images often contain data volumes far exceeding that of the single image, presenting significant challenges in storage and transmission for the overall system. By exploiting modality redundancy between different views, multi-view compression can achieve performance far beyond that of single-view compression, which has propelled multi-view compression to become one of the hot topics in recent years.

Early related researches primarily concentrated on traditional schemes. The Moving Picture Experts Group (MPEG) incorporated an extension into the HEVC coding standard specifically for efficiently encoding multi-view video and depth map data, known as 3D-HEVC [21]. 3D-HEVC employs inter-view prediction techniques, allowing video frames from one viewpoint to reference corresponding frames from other viewpoints for encoding. The method utilizes the correlation between different views to reduce modality redundancy significantly. In VVC [9], extensive works [20, 22] also leverage the additional information provided by depth maps to assist coding.

As deep learning technologies continue to evolve, researchers are increasingly focusing on end-to-end multi-view compression frameworks. The advantage of the framework lies in its ability to efficiently integrate information from different views, learning the intrinsic representation of data directly through neural networks, thereby achieving more efficient data compression [23, 30, 32, 33]. Compared to traditional methods that enhance performance by optimizing individual components, end-to-end approaches can achieve more significant performance improvements through global optimization. In the future, with further enhancements in computational resources and continuous optimization of algorithms, end-to-end multi-view compression holds promising application prospects.



**Figure 1: Rate-distortion curves for performance comparison tested in Instereo2K and KITTI dataset. It is worth noting that we calculate the average value on the left and right views.**

Despite the emergence of numerous new methods in the field, the diversity of training environments and test datasets poses severe challenges for conducting unified benchmark tests. CompressAI [5] is a well-known open-source algorithm library in the end-to-end image compression field, integrating several classic single-image compression frameworks [2, 3, 11, 19]. However, it is not compatible with multiple view images, which fails to address the dilemmas faced in the multi-view compression. This necessitates the urgent development of an open-source algorithm library capable of integrating efficient multi-view compression algorithms. In light of this, we established OpenMVC, an open-source algorithm library for learning-based multi-view compression. The library harmonizes nine classic algorithms in the field using the same paradigm. We provide detailed descriptions of each algorithm and analyze their core contributions. Additionally, we have conducted extensive benchmark testing of multiple algorithms across various metrics (Rate-Distortion performance and time complexity), filling the gap in the field.

In conclusion, our contributions can be summarized as follows:

- As far as we know, OpenMVC is the first open-source library specifically dedicated to learning-based multi-view compression, thereby bridging a gap in the field.
- We create identical training environments for several classic algorithms and regulate the testing conditions, which enhances the clarity and specificity of algorithm comparisons and categorizations in the field.
- We standardize the pipeline, interfaces, and common operators for end-to-end multi-view compression tasks. Various algorithms are integrated into a unified scheme to facilitate practical deployment and application.
- We conduct uniform benchmark tests for each algorithm under multiple metrics, including RD (Rate-Distortion) performance and encoding/decoding time.

## 2 Supported Algorithms

*HESIC*. *HESIC* [13] is one of the early and solid works in the field. It utilizes a deep regression model to estimate the homography matrix, which is applied to transform the left image. Residuals between

**Table 1: Quantitative RD performace comparisons for BD-PSNR (dB), BD-MSSSIM, and their corresponding BD-rate (%) on Instereo2K and KITTI dataset. The results in the table are obtained by comparing other methods with HESIC. Bolded text indicates that the method achieves optimal performance under the corresponding test condition.**

| Dataset    | Metric         | Method    |               |         |        |          |         |               |          |
|------------|----------------|-----------|---------------|---------|--------|----------|---------|---------------|----------|
|            |                | HESIC-GMM | BCSIC         | SASIC   | LDMIC  | LDMIC-CB | MASIC   | BiSIC         | BiSIC-CB |
| Instereo2K | BD-PSNR (dB) ↑ | 0.6072    | 1.9021        | 0.9861  | 1.8671 | 1.8026   | 0.1902  | <b>2.7465</b> | 2.4944   |
|            | BD-Rate (%) ↓  | -24.18    | -57.06        | -32.63  | -55.31 | -53.78   | -6.62   | <b>-73.92</b> | -69.36   |
|            | BD-MSSSIM ↑    | 0.0041    | 0.0074        | 0.0035  | 0.0077 | 0.0106   | 0.0017  | <b>0.0156</b> | 0.0135   |
|            | BD-Rate (%) ↓  | -20.44    | -33.46        | -14.97  | -34.07 | -46.60   | -6.64   | <b>-62.64</b> | -55.41   |
| KITTI      | BD-PSNR (dB) ↑ | 0.3229    | <b>3.1063</b> | 0.0203  | 1.3019 | 0.7661   | -0.2783 | 1.7682        | 1.6627   |
|            | BD-Rate (%) ↓  | -12.08    | <b>-56.61</b> | -0.74   | -36.74 | -23.81   | 11.94   | -49.98        | -47.22   |
|            | BD-MSSSIM ↑    | 0.0028    | 0.0072        | -0.0079 | 0.0045 | 0.0077   | 0.0021  | <b>0.0123</b> | 0.0098   |
|            | BD-Rate (%) ↓  | -11.04    | -26.62        | 35.84   | -16.07 | -27.18   | -7.62   | <b>-40.72</b> | -33.74   |

the left and right views are encoded to save bitrates. For entropy coding, context-based entropy models are used to eliminate redundant information between the stereo views. During the decoding process, image quality is improved by cross-quality enhancement. HESIC demonstrates substantial performance improvements over both traditional and deep learning-based single image compression frameworks.

*HESIC-GMM.* HESIC-GMM [13] is a more efficient compression framework derived from HESIC. The main difference between HESIC-GMM and HESIC lies in the entropy model. HESIC uses an autoregressive context entropy model, which results in slower encoding and decoding speeds, making it less suitable for practical deployment. To address this issue, HESIC-GMM employs a conditional entropy model based on Gaussian Mixture Model (GMM) to more accurately estimate the probabilities of latent representations. Compared to HESIC, HESIC-GMM significantly reduces time complexity.

*BCSIC.* Previous multi-view image coding methods typically contain clear dependency on different views. In contrast, BCSIC [17] enables mutual reference between different views, reducing inter-view redundancy. Additionally, cross-view prior information is used to guide the conditional entropy model to more accurately estimate the probability distribution of latent representations in each view, thereby further enhancing coding efficiency.

*SASIC.* Unlike the above methods, SASIC [25] leverages the similarity of overlapping fields in stereo image pairs to further assist multi-view reconstruction. By translating the latent representation of the left view through shifting to obtain the right view, it encodes the residual information of the right view to reduce overall bitrate costs significantly.

*LDMIC.* Existing multi-view coding schemes typically perform joint encoding from different views to enhance compression performance, relying on efficient collaboration among different cameras. In contrast, LDMIC [31] decoupling in the encoding process enables the system to possess efficient distributed source coding capabilities. Through joint decoding, LDMIC effectively captures similar features of different views for efficient reconstruction.

*LDMIC-CB.* Despite LDMIC is designed as an excellent distributed source coding framework, the autoregressive context entropy model results in slower encoding and decoding speeds, hindering practical deployment. Therefore, LDMIC-CB [31] is proposed to solve the issue. The main difference between LDMIC-CB and LDMIC is the use of a more efficient checkerboard context entropy model, which allows for efficient parallel encoding and decoding processes, significantly reducing time complexity.

*MASIC.* MASIC [12] is a derivative work of HESIC-GMM. MASIC introduces a mask prediction module that generates a guiding mask to direct the encoding and decoding process, addressing the issue of erroneous guidance in stereo image pairs. Additionally, MASIC incorporates a new training set called Palace, which includes hundreds of high-resolution stereo image pairs, significantly enhancing compression capability.

*BiSIC.* BiSIC [18] addresses compression imbalance via a symmetric bidirectional framework. Its 3D convolution backbone captures intra/inter-view correlations, while bidirectional mutual attention blocks enable cross-view global feature interaction. A novel cross-dimensional entropy model fuses hyperpriors, spatial/channel context, and stereo dependency for accurate latent distribution estimation.

*BiSIC-CB.* BiSIC-CB [18] addresses the computational overhead of BiSIC by introducing a stereo-checkerboard structure, which partitions latent representations into anchor and non-anchor regions for two-stage processing. This design replaces spatial autoregressive inference with parallelized 3D convolution-based context modeling, reducing codec time while preserving compression efficiency.

### 3 Benchmarking Results and Analysis

#### 3.1 Datasets

*3.1.1 InStereo2K.* InStereo2K dataset [4] comprises 2,050 pairs of stereo images captured from various scenes and environments. The images in the dataset include both indoor and outdoor scenes, covering various lighting conditions and object surface materials, thus presenting challenges for compression. In this experiment, we

randomly select 2,000 pairs for training, 25 pairs for validation, and the remaining 25 pairs for benchmark testing.

**3.1.2 KITTI.** KITTI dataset [14] is one of the most widely used and essential datasets in autonomous driving. In this study, we randomly choose more than 1,500 pairs from KITTI stereo image dataset for the training set, select 25 pairs for the validation set, and use the remaining 25 pairs for benchmark testing.

### 3.2 Evaluation Metrics

PSNR and MS-SSIM [24] serve as evaluation metrics to assess the RD performance during testing. PSNR measures image fidelity by quantifying pixel-wise differences through signal-to-noise ratio calculations, while MS-SSIM assesses structural similarity by modeling human visual perception of textures and edges. Additionally, we compared BD-Rate [7], BD-PSNR [8], and BD-MSSIM to obtain quantitative rate-distortion performance.

### 3.3 Rate-Distortion Performance Evaluation

We plot the RD curves of all algorithms in OpenMVC on Instereo2K and KITTI datasets to show the performance differences of each algorithm, as shown in Fig. 1. It is evident that the compression performance of each algorithm is better on Instereo2K, due to the fact that the KITTI dataset typically contains more complex scenes, making it difficult to fully utilize the additional information from different views. To further quantitatively compare the performance of each algorithm, we utilize BD-PSNR, BD-MSSIM and calculate BD-Rate using PSNR and MS-SSIM to determine the RD performance gain of each algorithm compared to HESIC. Based on RD performance comparisons across methods on Instereo2K and KITTI, BiSIC achieves optimal BD-PSNR, BD-MSSIM, and BD-rate on Instereo2K, primarily due to its cross-dimensional entropy module fully exploiting view redundancy to enhance inter-view information utilization. BCSIC excels on KITTI with the highest BD-PSNR and BD-rate, likely benefiting from its efficient structure in capturing view correlations. These results collectively highlight that leveraging attention-based modules for inter-view information sharing and adopting redundancy-exploitation mechanisms are critical for optimizing multi-view compression performance. Methods like MASIC and HESIC-GMM, lacking such designs, underperform significantly in most metrics, particularly on Instereo2K.

### 3.4 Running Time Evaluation

We compare the encoding and decoding time of several algorithms, and the results are shown in Table 2. The hardware setup includes an Intel(R) Xeon(R) Silver 4210R CPU and an NVIDIA GeForce RTX A5000 GPU workstation. During testing, we calculate the average encoding and decoding time across multiple bitrate points, from high to low. It can be observed that due to higher image resolution and content complexity, the encoding and decoding time for all methods are significantly longer in KITTI dataset. Methods using autoregressive context entropy models (such as HESIC, BCSIC, LDMIC) show notably longer encoding and decoding time, posing challenges for deployment under bandwidth-limited conditions. Despite not using autoregressive context entropy models, HESIC-GMM and MASIC introduce additional homography matrix

**Table 2: Average encoding time and decoding time comparison of different methods.**

| Model     | Instereo2K |        |        | KITTI  |        |        |
|-----------|------------|--------|--------|--------|--------|--------|
|           | Enc(s)     | Dec(s) | Tot(s) | Enc(s) | Dec(s) | Tot(s) |
| HESIC     | 6.829      | 10.991 | 17.820 | 10.279 | 16.611 | 26.890 |
| HESIC-GMM | 0.558      | 0.565  | 1.123  | 1.237  | 3.141  | 4.378  |
| BCSIC     | 10.481     | 15.731 | 26.212 | 15.543 | 23.888 | 39.431 |
| SASIC     | 0.097      | 0.096  | 0.193  | 0.125  | 0.130  | 0.255  |
| LDMIC     | 6.900      | 11.050 | 17.950 | 10.312 | 16.435 | 26.747 |
| LDMIC-CB  | 0.168      | 0.155  | 0.323  | 0.199  | 0.182  | 0.381  |
| MASIC     | 0.745      | 0.742  | 1.487  | 1.495  | 1.539  | 3.034  |

\* BiSIC and BiSIC-CB only use network inference and do not perform actual encoding and decoding, so the encoding and decoding time is not counted.

estimation networks, thereby increasing time complexity to some extent.

## 4 Conclusion

In this paper, we establish the first open-source library for learning-based multi-view compression. We provide an overview of the algorithms used in the library and summarize the core contributions of each algorithm. Subsequently, we conduct performance benchmark tests on these algorithms. We analyze the compression performance and time complexity of each model across multiple metrics. We sincerely hope that the open-source algorithm library will advance the development of deep learning-based multi-view compression, better serving various 3D vision tasks.

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